Artificial intelligence based assessment of minimally invasive surgical skills using standardised objective metrics - A narrative review.

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Abstract

Many studies display significant heterogeneity in the reliability of artificial intelligence (AI) assessment of minimally invasive surgical (MIS) skills. Our objective is to investigate whether AI systems utilising standardised objective metrics (SOMs) as the basis of skill assessment can provide a clearer understanding of the current state of such technology. We systematically searched Medline, Embase, Scopus, CENTRAL and Web of Science from March 2023 to September 2023. Results were compiled as a narrative review. Twenty-four citations were analysed. Overall accuracy of AI systems in predicting overall SOM score of a procedure ranged from 63 % to 100 %. The most frequently used SOM by AI algorithms were Objective Structured Assessment of Technical Skills (OSATS) (8/24) and Global Evaluative Assessment of Robotic Skills (GEARS) (8/24). Stratifying for AI studies which employed SOMs to assess surgical skill did not reduce heterogeneity of reported reliability. Our study identifies key issues within the current literature, which, once addressed, could allow more meaningful comparisons between studies.

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